

Tinghui Zhu

Email: darthzhu@gmail.com | [Personal Website](#) | [Google Scholar](#)

Research Interests

Self-evolving language and multimodal agents; long-horizon reasoning; learning from interaction and verifiable feedback.

Education

University of California, Davis	2025/09 – present
<i>Ph.D. in Computer Science.</i> Advisor: Muhao Chen	Davis, CA
Fudan University	2022/09 – 2025/06
<i>M.S. in Computer Science.</i> Advisor: Yanghua Xiao	Shanghai, China
Fudan University	2018/09 – 2022/06
<i>B.S. in Computer Science.</i>	Shanghai, China

Experience

Google DeepMind	2026/06 – present
<i>Software Engineering Intern, Mentor: Wenxuan Zhou</i>	Mountain View, CA
- Developed an online memory-management loop for long-horizon agents using counterfactual credit assignment to determine how memories should be updated; implemented a Git-native store that makes memory changes attributable and reversible. - Evaluated the system across five model families and three long-horizon agent benchmarks; active memory management mitigated regressions from unmanaged memory and improved agent performance by up to 17.8%.	
University of California, Davis, LUKA Group	2024/04 – 2025/06
<i>Research Intern, Mentor: Muhao Chen</i>	
- Studied cross-modality parametric knowledge conflicts in large vision-language models, developing a pipeline for conflict detection, interpretation, and inference-time mitigation. - Investigated modality extension and model merging for omni-modal models, analyzing their effects on language capability, cross-modal transfer, and generalization.	
The Ohio State University, OSU NLP Group	2023/07 – 2024/02
<i>Research Intern, Mentor: Yu Su</i>	
Proposed Deductive Beam Search, integrating beam search with deductive reasoning to reduce error accumulation in LLMs. Developed a bootstrapping method for synthesizing erroneous reasoning steps and validated the approach across diverse model scales. [COLM 2024]	

- Co-developed *TravelPlanner*, a tool-use planning benchmark with 1,225 curated planning intents and a sandbox over approximately four million records for evaluating real-world planning under multiple constraints. [ICML 2024 Spotlight]

Meituan, NLU group

2022/06 – 2023/03

Research Intern, Mentor: Rui Xie

- Developed a visual taxonomy expansion framework that integrates visual features into conceptual representations through prototype learning and cross-modal constraints, improving accuracy by 8.75% on the Chinese taxonomy benchmark. [ACMMM 2023]
- Built an automated taxonomy expansion pipeline that predicts hypernym relations between queries and clicked items from user interaction data.

Projects

Early Experience Scaling

2026/01 – 2026/06

- Built and released a 696K-example training corpus spanning eight embodied, tool-use, planning, and web-agent environments, supporting imitation learning, self-reflection, and implicit world modeling.
- Developed reproducible fine-tuning and evaluation infrastructure; reproduced consistent improvements over imitation learning across seven evaluated environments, with gains of up to 16.4 percentage points.

Publications

* Indicates equal contributions.

1. **Tinghui Zhu**, Sheng Zhang, James Y Huang, Selena Song, Xiaofei Wen, Yuankai Li, Hoifung Poon, Muhao Chen. Video Models Can Reason with Verifiable Rewards. *Under Review*, 2026.
2. Xiaofei Wen, Wenjie Jacky Mo, Xingyu Fu, Rui Cai, **Tinghui Zhu**, Wendi Li, Yanan Xie, Muhao Chen, Peng Qi. When Vision Speaks for Sound. *Under Review*, 2026.
3. Yuankai Li, **Tinghui Zhu**, Ha Min Son, Zhe Zhao, Xin Liu, Muhao Chen. AquaUI: Visual Token Reduction for GUI Agents with Adaptive Quadrees. *Under Review*, 2026.
4. Yixu Huang, **Tinghui Zhu**, Muhao Chen. Learning Adaptive Reasoning Paths for Efficient Visual Reasoning. *Under Review*, 2026.
5. Boyu Zhu, Xiaofei Wen, Wenjie Jacky Mo, **Tinghui Zhu**, Yanan Xie, Peng Qi, Muhao Chen. OmniGuard: Unified Omni-Modal Guardrails with Deliberate Reasoning. *Under Review*, 2025.
6. James Y Huang, Sheng Zhang, Qianchu Liu, Guanghui Qin, **Tinghui Zhu**, Tristan Naumann, Muhao Chen, Hoifung Poon. Be My Eyes: Extending Large Language Models to New Modalities Through Multi-Agent Collaboration. *Under Review*, 2025.
7. **Tinghui Zhu***, Kai Zhang*, Muhao Chen, Yu Su. Is Extending Modality The Right Path Towards Omni-Modality?. *ICLR 2026 Workshop on Multimodal Intelligence (ICLR Workshop)*, 2026.
8. Sirui Xia, Aili Chen, Xintao Wang, **Tinghui Zhu**, Yikai Zhang, Jiangjie Chen, Yanghua Xiao. Can LLMs Learn to Map the World from Local Descriptions?. *Under Review*, 2025.

9. Caiyu Hu, Yikai Zhang, **Tinghui Zhu**, Yiwei Ye, Yanghua Xiao. MCiteBench: A Multimodal Benchmark for Generating Text with Citations. In *Proceedings of the 30th Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.
10. **Tinghui Zhu**, Qin Liu, Fei Wang, Zhengzhong Tu, Muhao Chen. Unraveling Cross-Modality Knowledge Conflicts in Large Vision-Language Models. *Under Review*, 2024.
11. **Tinghui Zhu***, Kai Zhang*, Jian Xie, Yu Su. Deductive Beam Search: Decoding Deducible Rationale for Chain-of-Thought Reasoning. In *Proceedings of the 1st Conference on Language Modeling (COLM)*, 2024.
12. Siye Wu*, Jian Xie*, Jiangjie Chen, **Tinghui Zhu**, Kai Zhang, Yanghua Xiao. How Easily do Irrelevant Inputs Skew the Responses of Large Language Models?. In *Proceedings of the 1st Conference on Language Modeling (COLM)*, 2024.
13. Jiangjie Chen, Xintao Wang, Rui Xu, Siyu Yuan, Yikai Zhang, Wei Shi, Jian Xie, Shuang Li, Ruihan Yang, **Tinghui Zhu**, Aili Chen, Nianqi Li, Lida Chen, Caiyu Hu, Siye Wu, Scott Ren, Ziquan Fu, Yanghua Xiao. From Persona to Personalization: A Survey on Role-Playing Language Agents. *Transactions on Machine Learning Research (TMLR)*, 2024.
14. Jian Xie*, Kai Zhang*, Jiangjie Chen, **Tinghui Zhu**, Renze Lou, Yuandong Tian, Yanghua Xiao, Yu Su. TravelPlanner: A Benchmark for Real-World Planning with Language Agents, In *Proceedings of the 42nd International Conference on Machine Learning (ICML Spotlight)*, 2024.
15. **Tinghui Zhu**, Jingping Liu, Jiaqing Liang, Haiyun Jiang, Yanghua Xiao, Zongyu Wang, Rui Xie, Yunsen Xian. Towards Visual Taxonomy Expansion. In *Proceedings of the 32nd ACM Multimedia Conference (ACMMM)*, 2023.
16. Chao Wang*, **Tinghui Zhu***, Zhixu Li, Jingping Liu. SLR: A Million-Scale Comprehensive Crossword Dataset for Simultaneous Learning and Reasoning. **Neurocomputing**, 2023.
17. Lihan Chen, **Tinghui Zhu**, Jingping Liu, Jiaqing Liang, Yanghua Xiao. End-to-end Entity Linking with Hierarchical Reinforcement Learning. In *Proceedings of the 37th Annual AAAI Conference on Artificial Intelligence (AAAI)*, 2023.

Awards

- **Graduate Fellowship**, University of California, Davis 2025/09
- **National Scholarship** (top 1%), Fudan University 2023/10

Services

Conference reviewer: ACL (2023–2026); COLM (2024–2026); EMNLP (2024–2026); ICLR (2025–2026); NeurIPS (2026).